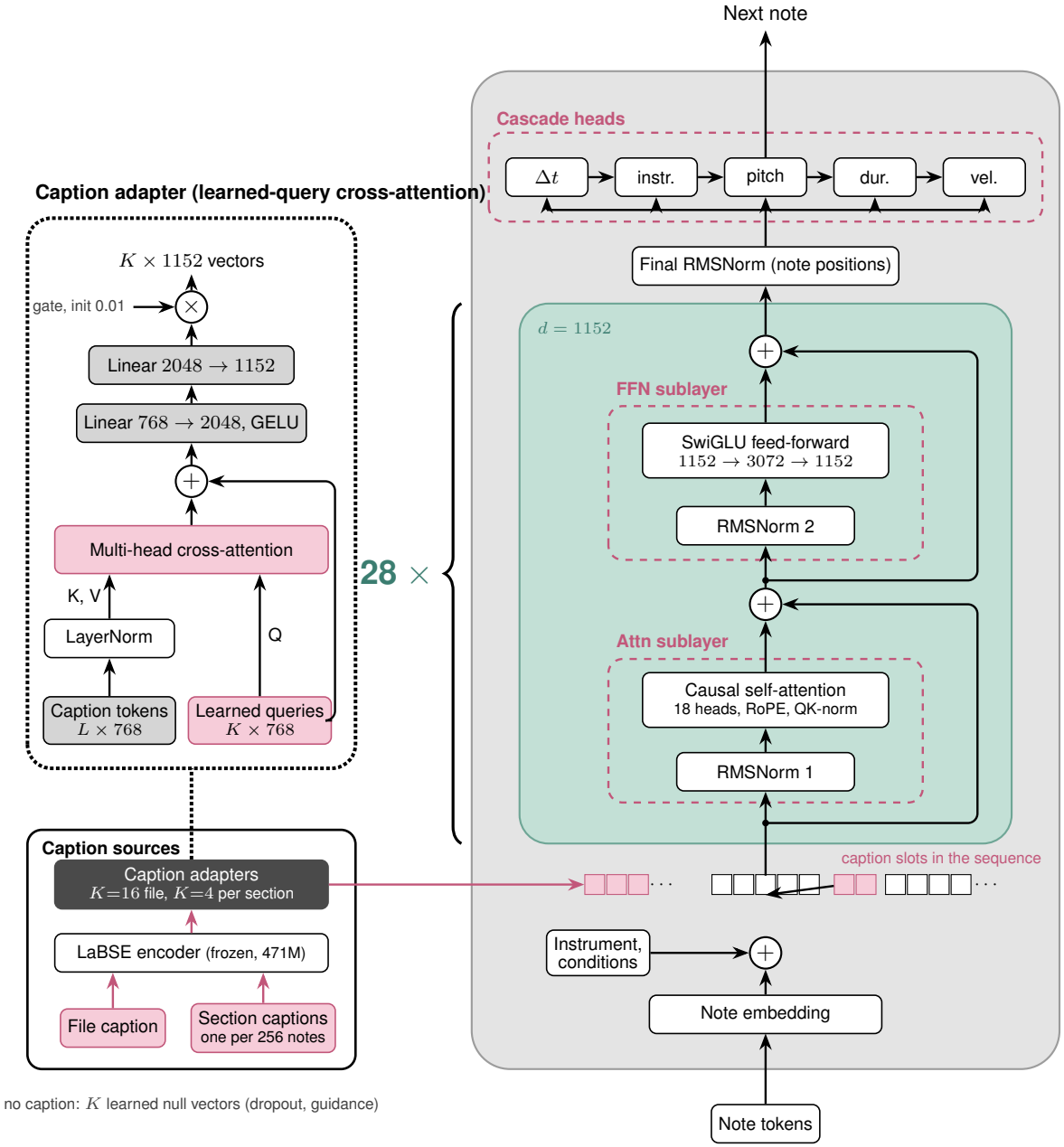
Captioned music model (477M)



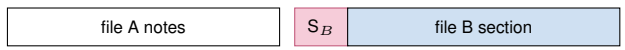
sample input: 2,048 notes + "loud rock, fast"

Keeping section captions in play (training)

1. Change-aware captions

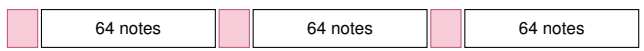
Each section caption names what changed vs the section before ("drums enter, louder"); half of the windows sit on a section change.

2. Spliced windows (25%)



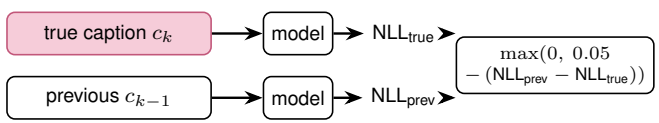
File A up to a section boundary, then a section of file B with B's caption and no file caption: the caption contradicts the context and has to win.

3. Caption repeated every 64 notes



The same 4 section vectors are inserted again, so the caption is never far back in the context.

4. Counterfactual loss + recognition head



On the first 32 notes of a changed section the true caption must beat the previous one by 0.05 nats (weight 1.0); a linear head on those notes' states tells true from previous (BCE, weight 0.1). Training only.

5. Dropout → guidance

File captions dropped 15%, section captions 20% (null vectors), so generation can use classifier-free guidance, scale 2.

F1 section recipe

```
section length      : 256 notes, 4 vectors
change windows      : 50%
spliced windows     : 25%
caption repeat      : every 64 notes
counterfactual loss : weight 1.0, margin 0.05
recognition head    : weight 0.1
switch share @1,024 : 0.01 (v2_A0) → 0.85 (all)
```